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The effects of different neck training methods on the neck function of aviation cadets

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To compare the effects of different neck eccentric training devices on the neck strength and endurance of aviation cadets, and to explore their roles in the prevention of neck injuries, providing a basis for introducing specialized neck training in adolescent aviation schools. Two intervention groups used a helmet-style neck training device and elastic bands for neck training, respectively, while the control group underwent regular resistance physical training without additional neck intervention. Tests were conducted on neck flexion, extension, lateral flexion, and rotation in six directions before the intervention, and after 6 and 12 weeks. Significant time and group interaction effects were found in all six directions of neck strength indicators (F-values were 4.834, 8.496, 10.359, 6.849, 3.324, 2.405, p<0.05). The neck strength in all six directions significantly increased for both intervention groups at 6 and 12 weeks (P<0.05). In the control group, a significant increase in strength was observed in all directions except for extension at 6 weeks (P<0.05), but no significant changes were observed after 6 weeks (P>0.05). A significant time and group interaction effect was also found for endurance indicators (F=6.204, P<0.01). All three groups showed a significant increase in neck endurance at 6 weeks, but no significant changes were observed in the elastic band group and the control group after 6 weeks (P>0.05); the helmet group showed significantly higher neck endurance at 12 weeks (103.55 ±47.68) seconds, (P<0.01). Both 12-week helmet-style neck training and elastic band neck training can enhance trainee neck strength and flexural endurance, with the helmet-style training showing a more significant improvement in neck endurance.

Keywords Aviation cadets, Neck training, Eccentric training, Strength, Endurance

With the advancement of modern fighter jet technology, neck injuries have become prevalent among military pilots. Studies show that the incidence of neck pain among pilots reaches over 80% within a year[1–3]. Neck pain is not merely a matter of physical discomfort; it can also have a negative impact on pilots' mental health, affecting their attention and concentration. This, in turn, reduces their ability to control the aircraft, leads to unstable postures, and in some cases, prevents them from performing essential flight maneuvers. Such issues may not only result in mission interruptions but could also lead to temporary or permanent grounding of the pilots[4], the consequences brought about by this impact are very serious. Honkanen et al.'s research shows that the pressure on pilots' neck muscles increases linearly with the increase in load. When the neck is in a non-neutral position, especially when pilots perform head-turning movements under high G loads, the neck pressure reaches its maximum[5]. Therefore,Neck injuries have severely impacted the completion of flight training and missions, becoming the leading musculoskeletal disorder causing pilots to be grounded[6,7]. Various countries have proposed specialized neck training for pilots to enhance neck strength and endurance, aiming to prevent neck injuries. Research shows that neck-specific training is very important for improving the neck strength of fighter pilots and preventing neck injuries[8]. Although there is ongoing exploration of neck training regimens, there is still no evidence-based training protocol specifically for the prevention of neck injuries among pilots[9]. Extensive research has already confirmed the positive effects of eccentric training in injury prevention[10], The core principle lies in the muscle being lengthened while contracting, playing a crucial role in braking, deceleration, and motion control. Compared to concentric contraction, eccentric contraction can generate greater tension and provides deeper stimulation to neuromuscular control and connective tissues. Therefore, it is widely applied in rehabilitation, yet research on its application in neck training remains relatively scarce.

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The Youth Aviation School is a crucial stage in China's efforts to identify and cultivate future military pilots at an early stage. Students admitted to this school are rigorously selected adolescents who are at a critical period of physical fitness and musculoskeletal development. The training at this stage aims to lay a solid physical foundation for their subsequent entry into advanced flight academies and ultimately becoming high-performance fighter pilots. Introducing the physical qualities required for the aviation profession can lay the foundation for anti-G fitness[11]. Adolescents have significant room for improvement in their physical constitution and body shape. Through the correction of physical constitution and body shape during adolescence, there is a promising prospect for the improvement of their anti-load capacity in the future[12].Studies have shown that the muscle growth rate is fastest in the human body between the ages of 15 and 18, and by the age of 18, muscles are nearly at the adult level. Cadets at the youth aviation school are in the golden period of muscle growth. Shifting the anti-G physical training, which focuses on increasing neck muscle strength, to the youth aviation school stage can greatly improve training efficiency and is of great significance for enhancing the anti-G capability of future military pilots in our air force[13].Therefore, this study is the first to introduce specialized neck training into youth aviation schools, using different types of neck training equipment for eccentric training, investigating their effects on cadets' neck strength and endurance to provide a basis for specialized neck training in this population.

Based on the above background, this study proposes the following hypotheses: Compared with the control group undergoing conventional training, both 12-week helmet-mounted cervical eccentric training and elastic band cervical eccentric training can significantly improve trainees' strength in six directions of neck flexion, extension, lateral flexion, and rotation, as well as neck flexor endurance. Moreover, since the helmet-mounted device can provide more direct and controllable load, its effect on improving neck endurance will be superior to that of elastic band training.

Methods and subjects

Subjects

The present study was conducted from April to July 2021 using a stratified random design on 112 male cadets from a youth aviation academy. To avoid inter-group interference within the same class, two senior classes were designated as intervention cohorts: one randomly assigned to helmet-based equipment training and the other to resistance band training; concurrently, 30 junior cadets served as the control group. Due to pandemic-related restrictions, physical training for senior and junior students was conducted simultaneously. All subjects followed the standard anti-gravity physical training curriculum for youth aviation academies without additional neck-specific interventions. Junior students exhibited lower baseline levels in conventional anti-gravity fitness training, with both comprehensive neck strength and endurance being inferior to those of senior students, thereby rendering the effects of the standard training more pronounced in this cohort. This research aims to evaluate the additional value of specialized neck training compared to conventional training. Due to the nature of the intervention (physical training), blinding the subjects and coaches was not feasible. However, all outcome assessments were conducted by a research assistant unaware of group allocation to minimize measurement bias.After excluding cadets who did not complete the three tests and training, the final effective data analyzed included 95 people: 33 in the helmet group, 32 in the elastic band group, and 30 in the control group. Basic information is provided in Table 1. Except for the age of the junior control group, there were no statistically significant differences among the other groups (P>0.05). This study was approved by the Ethics Committee of Beijing Sport University (Approval No: 2019113H), and all participants and guardians signed informed consent forms.

Intervention protocol

Both intervention groups undertook neck training sessions lasting approximately 25 min, three times a week. In each session, subjects first carried out 3 min of deep cervical flexor activation to prevent injuries, followed by 20 min of neck training, and finally, 2 min of neck stretching. The intervention training was conducted in the school's indoor laboratory and was supervised jointly by coaches and researchers. Each group of participants was not disturbed by other people during the training. The coaches and researchers only provided verbal instructions on the training content and the timing between groups. No additional influences were exerted during the training process.

The equipment used for neck training differed between the two intervention groups. The helmet group used helmet-type load devices developed specifically for military pilot neck training, while the elastic band group used Thera-Band equipment (U.S.A.). Both groups followed identical training content as outlined in Table 2, with a generally consistent increase in intensity. The helmet group gradually increased the load up to 3 kg, whereas the elastic band group progressed from blue bands to gold bands.

Progressive loading standards: training load is assessed weekly. When a trainee can effortlessly complete all sets and repetitions with proper form in a given exercise, and the Rating of Perceived Exertion (RPE) falls below "somewhat hard," the load should be increased in the following week's training. The helmet group increments

Group	N	Height (cm)	Weight (kg)	BMI (kg/m^2)	Age (years)
Helmet group	33	176.09 ± 4.64	68.06 ± 6.88	21.95 ± 2.04	16.61 ± 0.50
Elastic band group	32	176.09 ± 4.90	68.48 ± 7.03	22.71 ± 2.31	16.69 ± 0.47
Control group	30	173.82 ± 6.15	65.06 ± 6.62	21.51 ± 1.58	15.97 ± 0.32

Table 1. Basic information of subjects (M±SD).

Exercise content	Description
Supine neck lift training	Lie supine on a mat with knees bent, tuck in the chin, lift head quickly, and lower slowly (5-s duration). Perform 10 repetitions, rest for 30 s between 2 sets
Right side-lying training	Lie on the right side on a mat, support on one elbow and knee, tuck in the chin, quickly flex head to the left (as fast as possible), lower head slowly (5 s). Perform 10 repetitions, rest for 30 s between 2 sets
Left side-lying training	Symmetrical exercise to right side-lying training
Quadruped neck lift training	Start in a hands-and-knees position on a mat, tuck in the chin, lift head quickly, and lower slowly (5 s). Perform 10 repetitions, rest for 30 s between 2 sets
Supine neck lift and rotation	Lie supine on a mat with knees bent, tuck in the chin, lift head quickly, rotate to the left and return slowly (5 s), rotate to the right and return slowly (5 s). Perform 10 repetitions, rest for 30 s between 2 sets
Quadruped neck lift and rotation	Start in a hands-and-knees position on a mat, tuck in the chin, lift head quickly, rotate to the left and return slowly (5 s), rotate to the right and return slowly (5 s). Perform 10 repetitions, rest for 30 s between 2 sets

Table 2. Neck strength training content.

weights in 0.5kg steps up to 3kg; the resistance band group progresses sequentially through Thera-Band resistance levels, advancing from blue bands to black, silver, and ultimately gold bands.

Test metrics

(1) Isometric neck strength

A handheld dynamometer, Micro-Fit3 Hoggan (USA), was used to measure the strength of neck flexion, extension, lateral flexion (left and right), and rotation (left and right). The units are in Newtons (N). Before the commencement of daily testing, the device undergoes zero calibration in accordance with the manufacturer's guidelines. Neck flexion, extension, and lateral flexion were measured while the subject was in a seated position, while rotation was measured in the supine position. During the test, the subject exerted force continuously for 5 seconds. Two tests were completed in each direction, and the average value was taken. There was a 10-second interval between each test and a 30-second interval when changing directions.

(2) Neck endurance

A helmet-type neck device was used for testing, with an additional 2 kg of weights attached. The subject assumed a supine position, with both legs bent and hands placed on the abdomen. The subject was instructed to maintain a neutral neck position and keep their shoulders flush with the contact surface. They were then required to perform a 30° forward neck flexion and maintain this position until their head dropped, marking the end of the test. The time sustained in this position was recorded, with units in seconds (s).

Data analysis

The SPSS 26.0 software was used for statistical analysis of the data. Normally-distributed metric data were represented as $M \pm \mathrm{SD}$. Repeated measures ANOVA was employed to examine data from the three groups across three measurements. When Mauchly's test indicated $\mathrm{P} < 0.05$, GreenhouseGeisser correction was applied. If the interaction effects were statistically significant, further simple effects analyses were carried out. When interaction effects were not significant but the main effect was significant, pairwise comparisons between time points within the group were conducted using the Bonferroni method. The significance level was set at $\mathrm{P} < 0.05$.

Research results

Changes in neck strength

The description of various indicators for neck strength is given in Table 3. After conducting a spherical test, both the main effect of time and the interaction effect between time and group were found to be significant. The interaction effect for right rotation strength was marginally significant. After performing simple effect analysis, the contour graph for time and group interaction is shown in Fig. 1.

Changes in neck endurance

The description of neck endurance indicators is shown in Table 4. Repeated measures ANOVA revealed a significant main effect of time ($\mathrm{F} = 47.417, \mathrm{P} < 0.01$), suggesting differences in neck endurance across time points. There was also a significant interaction effect between time and group ($\mathrm{F} = 6.204, \mathrm{P} < 0.01$), indicating that the trend in endurance changes varied across the three groups. The cohens'd values for the helmet group were 0.84 (at week 6) and 1.01 (at week 12), while for the elastic band group, they were 0.68 (Week 6) and 0.30 (Week 12). Simple effect analysis is shown in Fig. 2.

Discussion

There are numerous reasons that can lead to neck injuries in pilots, such as the load of personal protective equipment, head movements during flight, and various factors like the action of loads. Among them, high overload and cumulative overload can cause changes in the stress state of the cervical spine, significantly increasing the load-bearing stress. This, in turn, affects the soft tissues, vertebrae, and intervertebral discs, and in severe cases, may even lead to pathological changes[14]. Military pilots' neck injuries are related to repeated exposure to high G-forces[10,15]. 70% of high-performance fighter pilots in the UK report flight-related neck pain[16]. 48% of Finnish pilots have experienced acute neck pain during missions[17]. During flights, the load can

Metric	Group	0 weeks	6 weeks	12 weeks	d(Wee6)	d(Week12)	F-value	P-value
Flexion	Control (n = 30)	40.27 ± 6.71	52.18 ± 9.00*	53.59 ± 10.40#				-
	Helmet (n = 33)	44.9 ± 8.86	61.72 ± 10.52*	70.57 ± 11.02*#	0.97	1.58	4.834	0.003
	Elastic band (n = 32)	42.23 ± 9.23	57.68 ± 8.46*	64.43 ± 11.47*#	0.63	0.99		-
Extension	Control (n = 30)	46.15 ± 6.64	54.05 ± 8.88	52.14 ± 9.50#				
	Helmet (n = 33)	60.16 ± 9.25#	63.4 ± 14.90*	74.43 ± 13.42*	0.75	1.90	8.496	<0.01
	Elastic band (n = 32)	51.1 ± 11.78	58.52 ± 13.01*	69.73 ± 12.11*	0.40	1.61		
Lateral flexion(left)	Control (n = 30)	36.45 ± 5.07	46.99 ± 9.55*	43.67 ± 6.57#				
	Helmet (n = 33)	44.33 ± 9.75#	54.81 ± 9.37*#	62.14 ± 10.67*#	0.83	2.06	10.359	<0.01
	Elastic band (n = 32)	34.9 ± 6.89	46.4 ± 7.47*	52.83 ± 7.36*#	0.07	1.31		
Lateral flexion (right)	Control (n = 30)	36.76 ± 4.83	45.68 ± 6.17	43.38 ± 6.60#				
	Helmet (n = 33)	45.06 ± 9.47#	54.95 ± 9.20*#	59.82 ± 11.22*#	1.17	1.77	6.849	<0.01
	Elastic band (n = 32)	35.45 ± 7.67	46.93 ± 5.87*	51.33 ± 7.82*#	0.21	1.10		
Rotation (left)	Control (n = 30)	41.91 ± 6.60#	48.21 ± 9.04*#	49.57 ± 7.59#				
	Helmet (n = 33)	52.81 ± 11.38	64.95 ± 11.49*#	68.31 ± 11.34*#	1.61	1.92	3.324	0.012
	Elastic band (n = 32)	48.58 ± 8.07	56.32 ± 5.48*#	60.2 ± 8.53*#	1.09	1.31		
Rotation (right)	Control (n = 30)	40.79 ± 4.50#	52.41 ± 8.50*#	56.07 ± 6.50*#				
	Helmet (n = 33)	53.03 ± 11.64	68.7 ± 10.65*#	72.5 ± 8.31*#	1.68	2.19	2.405	0.051
	Elastic band (n = 32)	50.34 ± 7.96	62.29 ± 6.50*#	66.24 ± 8.05*#	1.31	1.39		

Table 3. Changes in neck strength (M ± SD). * Indicates a statistically significant difference within the group compared to the previous test ($P < 0.05$). # indicates a statistically significant difference between the groups ($P < 0.05$)

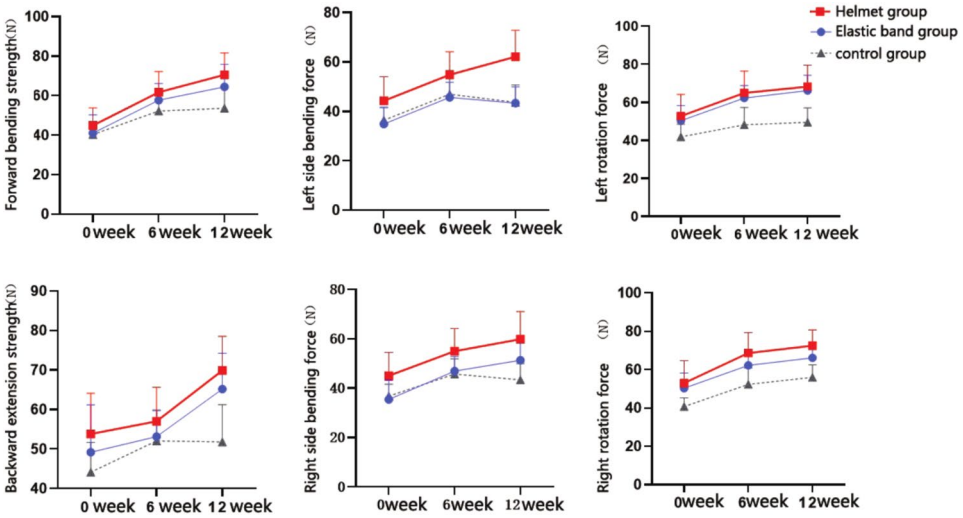


Fig. 1. Temporal and group interaction contour graph for neck strength.

Group	Week 0	Week 6	Week 12	d(Week 6)	d(Week 12)	F-value	P-value
Control (n = 30)	42.64 ± 21.66#	52.20 ± 21.59*	63.37 ± 27.97				
Helmet (n = 33)	55.81 ± 21.88	71.73 ± 24.85*	103.55 ± 47.68*#	0.84	1.01	6.204	< 0.01
Elastic Band (n = 32)	50.82 ± 17.29	69.68 ± 29.32*	71.31 ± 22.65	0.68	0.30		

Table 4. Statistical description of neck endurance indicators (M ± SD). *Indicates a statistically significant difference within the group compared to the previous test ($P < 0.05$). # indicates a statistically significant difference between the groups ($P < 0.05$)

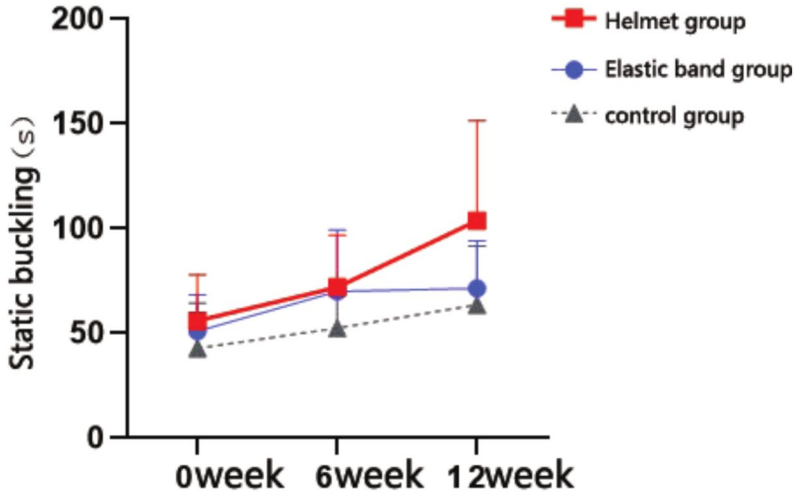


Fig. 2. Temporal and group interaction contour graph for neck endurance.

exceed 2G more than 20% of the time, reaching up to 9G. Pilots' helmets, due to additional equipment like night vision and data processing, become heavier and more front-heavy[18–20], thereby increasing the torque load in high G environments[21]. To maintain head functionality, neck muscles need to contract strongly to counteract these forces, leading to fatigue and potential acute or chronic injuries[22,23]. Proactive development of training protocols that effectively enhance cervical muscular strength and endurance in pilots is imperative to improve neck functional capacity, augment the efficacy of other flight training components, and ultimately optimize mission performance. Furthermore, Prevention and rehabilitation of neck pain are also crucial after training or in daily life[24].

Drury's report suggests that adolescence is a key period for training to improve physical conditioning, performance, and injury prevention[15], making it an opportune time for specialized training. During the flight of high-performance fighter jets, the anti-G maneuvers performed by pilots involve muscle groups throughout the body, including the calves, thighs, abdomen, chest, arms, etc. Pilots are required to rapidly tense their entire body muscles and maintain this tension for a certain period of time to counteract the rapid increase in G-forces generated by the high-performance fighter jets[25]. Therefore, during the critical period of growth and development, by integrating the characteristics of human development with the specific physical requirements of fighter pilots, and on the premise of effectively improving aerobic endurance, focusing on enhancing the strength of all body muscles–especially emphasizing the development of core and lower limb strength–can more effectively satisfy the physical demands imposed by high-G maneuvers. This is instrumental in establishing a robust physical foundation for elite fighter pilots and substantially enhancing their combat capability[26].

Our results show an increase in neck strength and endurance across all groups at the 6-week mark. After 6 weeks, the control group's improvement plateaued, while the two intervention groups continued to make gains. Notably, the helmet group saw an 85.5% improvement in endurance at 12 weeks, twice as effective as the band group, highlighting the helmet's potential for dramatically improving neck endurance.Therefore, For pilots whose flight training and mission performance are compromised by inadequate cervical strength and endurance, this targeted intervention measure can achieve the effect of preventing neck injuries. Compared with blind training, formulating a scientific and targeted special training plan is the only way to effectively improve the effectiveness of neck strength training. In an effective training plan, training methods, timing, cycles, loads, and sequences are all important factors that need attention[27].

The neck training program in this study is time-efficient, requiring simple and portable equipment, making it feasible for aviation trainees. The training program incorporates deep neck muscle activation, strength training, and stretching. Prior research has emphasized the critical role deep flexor muscles play in maintaining cervical posture, stability, and endurance, as well as their association with cervical injuries[28,29]. The present study particularly emphasizes the involvement of deep muscles in neck exercises. Each training session begins with activation exercises for these muscles, followed by neck strength training. Eccentric movement can enhance neuromuscular control and prevent sports injuries, and it has wide applications in competitive sports and rehabilitation. A study comparing concentric and eccentric neck training demonstrated that eccentric training has more pronounced effects on neck circumference, strength, and endurance[30].In this study, both intervention groups incorporated eccentric training components, and their strength change patterns were generally consistent, significantly higher than those of the control group. Regarding endurance indicators, the helmet group showed a significant improvement at 12 weeks. This may be because the neck flexion and rotation training in the helmet group were performed in a supine position, while the resistance band group's rotation training was performed in a seated position. As a result, the deep neck flexor muscles in the helmet group were more engaged during training, receiving greater training stimuli. Consequently, the helmet group experienced a substantial increase in neck flexion endurance at 12 weeks. Strengthening neck muscle strength training and maintaining the correct head posture during flight are currently effective measures to prevent and reduce neck injuries[31]. Pilots in countries such as the United States, Finland, Switzerland, and Australia attach great importance to

the strength training of neck muscles. Domestic scholars have also attempted to use various methods for neck muscle training[32].

Based on the 12-week intervention results with three sessions per week in this study, we propose the following prospects for subsequent training arrangements: To maintain and further enhance the functionality of neck muscles in the long term, it is necessary to institutionalize and regularize specialized neck training. A feasible strategy is to adjust the training frequency to one or two maintenance sessions per week after completing the high-intensity initial training phase. Such an arrangement not only aligns with the scientific principles of training periodization but also offers greater practical operability, effectively integrating neck training into the long-term physical training system of aviation cadets, thereby providing sustained protection for pilots' necks during flight.

In summary, both 12-week helmet-mounted and elastic band neck eccentric training can effectively improve trainees' neck function. This study provides promising prospects and important physiological evidence for improving pilots' cervical function and enhancing cervical training efficacy, thereby mitigating neck-related issues during flight operations; however, its specific prophylactic efficacy requires validation through longitudinal studies employing injury incidence as a direct endpoint and assessing changes in injury incidence.

Limitations & future work

This study investigated the efficacy of various neck training modalities on aviation cadets, yielding meaningful findings. However, several limitations should be acknowledged. First, the age disparity between the intervention and control groups may constitute a potential confounding variable, potentially influencing the outcomes. Second, the modest sample size (approximately 30 participants per group) may limit the generalizability of our findings[33–35]. Third, gender differences were not examined, as all participants were male.

Based on the above limitations, future research should: (1) utilize age-matched cohorts for intervention and control conditions to strengthen internal validity and minimize confounding effects attributable to age; (2) enroll larger and more heterogeneous samples (e.g., aviation cadets from multiple training institutions or geographic regions) to enhance external validity and broaden generalizability; (3) examine potential sex-specific training responses to establish whether protocols require gender-specific optimization; (4) prolong the intervention period to assess the long-term efficacy and sustainability of training adaptations; and (5) investigate novel and optimized training modalities to determine optimal efficacy.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request

Code availability

The code provided in this study is real and usable.

Materials availability

This study uses external physical testing devices, which will not produce side effects to people, and is safe and easy to implement.

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Author contributions

Lu hao is responsible for determining the research direction, formulating the research framework and writing the first draft; Zhao dingyu is responsible for the later modification of the article and statistical analysis data; Jia xueru is responsible for the experiment and data collection. The corresponding author received a submission invitation from this journal and was advised to submit the manuscript.

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Declarations

Ethics approval and consent to participate

This study was approved by the Sports Science Experimental Ethics Committee of Beijing Sport University, and the research methods were conducted in accordance with the approved guidelines. All participants provided informed consent prior to the experiment.

Competing interests

The authors declare no competing interests.

Additional information

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